

residents.)

### ***Grass and Trees***

I'll summarize here and repeat the arguments previously made throughout the book. Measures of uncertainty that are based on the bell curve simply disregard the possibility, and the impact, of sharp jumps or discontinuities and are, therefore, inapplicable in Extremistan. Using them is like focusing on the grass and missing out on the (gigantic) trees. Although unpredictable large deviations are rare, they cannot be dismissed as outliers because, cumulatively, their impact is so dramatic.

The traditional Gaussian way of looking at the world begins by focusing on the ordinary, and then deals with exceptions or so-called outliers as ancillaries. But there is a second way, which takes the exceptional as a starting point and treats the ordinary as subordinate.

I have emphasized that there are two varieties of randomness, qualitatively different, like air and water. One does not care about extremes; the other is severely impacted by them. One does not generate Black Swans; the other does. We cannot use the same techniques to discuss a gas as we would use with a liquid. And if we could, we wouldn't call the approach "an approximation." A gas does not "approximate" a liquid.

We can make good use of the Gaussian approach in variables for which there is a rational reason for the largest not to be too far away from the average. If there is gravity pulling numbers down, or if there are physical limitations preventing very large observations, we end up in Mediocristan. If there are strong forces of equilibrium bringing things back rather rapidly after conditions diverge from equilibrium, then again you can use the Gaussian approach. Otherwise, fuhgedaboudit. This is why much of economics is based on the notion of equilibrium: among other benefits, it allows you to treat economic phenomena as Gaussian.

Note that I am not telling you that the Mediocristan type of randomness does not allow for *some* extremes. But it tells you that they are so rare that they do not play a significant role in the total. The effect of such extremes is pitifully small and decreases as your population gets larger.

To be a little bit more technical here, if you have an assortment of giants and dwarfs, that is, observations several orders of magnitude apart, you could still be in Mediocristan. How? Assume you have a sample of one thousand people, with a large spectrum running from the dwarf to the giant. You are likely to see many giants in your sample, not a rare occasional one. Your average will not be impacted by the occasional additional giant because some